

Notizen Little Big Topo OGOSMO

☰ Tags	
🕒 Zuletzt bearbeitet am	7. April 2026 21:45
🕒 Erstellt am	12. Februar 2026 10:57

Todo

Aa Name	👤 Assign	⚙ Status
1.2 oslo-paw Jumphost aufsetzen	 Eduard Smola	Done
oslo-srv2 Dovecot	 Zwettler Benjamin	Done
AD CS fixen	 Eduard Smola	Done
th-srv KeadHCP	 Marcel	Done
oslo-srv1 rsyslog	 Marcel	Done
1.1 oslo-dc2 DFS	 Eduard Smola	Done
oslo-fw Inter VLAN Connectivity	 Eduard Smola	Done
1.3 oslo-dc1 Zertifikate per GPO	 Eduard Smola	Done
1.4 NPS auf Cisco C8000v	 Eduard Smola	Done
1.5 oslo-dc1 Adv. Audit Policies & Security Baseline	 Eduard Smola	Done
oslo-srv1 uFTPd	 Marcel	Done
oslo-srv1 Apache	 Marcel	Done
th-dc1 Replikationsfehler beheben	 Eduard Smola	Done
oslo-srv2 Postfix	 Zwettler Benjamin	Done
oslo-srv2 BIND9	 Zwettler Benjamin	Done
th-srv prometheus	 Marcel	Done
th-srv Grafana	 Marcel	Done

Github

<https://github.com/1059-25-HTL3R/schule>

VPN auf die UCS

Im FortiClient

Passwort für IPsec-VPN

Earpiece-Corny2-3scoop-Gwyn-123

Credentials für VPN-User

username Diplomanden

password earpiece-corny2-3scoop-Gwyn-Lush9-needy-iii7

The image shows the 'Edit VPN Connection' window in FortiClient. The 'IPsec VPN' tab is selected. The 'Authentication Method' is set to 'Pre-shared key'. A red handwritten note 'IPsec secret' is next to the pre-shared key field. The 'Authentication (XAuth)' section is set to 'Save login' with the username 'Diplomanden'. The 'Remote Gateway' is '194.50.115.124'. Below this, the 'VPN Settings' section is expanded, showing 'Phase 1' and 'Phase 2' configurations. In 'Phase 1', the 'Encryption' is 'AES128', 'Authentication' is 'SHA1', 'DH Group' is '5' (circled in red), 'Key Life' is '86400' sec, and 'Local ID' is '232413' (circled in red). 'Dead Peer Detection' and 'NAT Traversal' are checked. 'Phase 2' shows 'Encryption' as 'AES128', 'Authentication' as 'SHA1', 'Key Life' as '43200' seconds, and 'DH Group' as '5' (circled in red). 'Enable Replay Detection' and 'Enable Perfect Forward Secrecy (PFS)' are checked.

Edit VPN Connection

VPN: **SSL-VPN** | **IPsec VPN** | XML

Connection Name: UCS

Description:

Remote Gateway: 194.50.115.124 [X]

+ Add Remote Gateway

Authentication Method: Pre-shared key

IPsec secret

Authentication (XAuth): ☐ Prompt on login ☒ Save login ☐ Disable

Username: Diplomanden

Failover SSL VPN: [None]

Single Sign On Settings: ☐ Enable Single Sign On (SSO) for VPN Tunnel

VPN Settings

Phase 1

IKE Proposal: Encryption: AES128 Authentication: SHA1

Encryption: AES256 Authentication: SHA256

DH Group: ☐ 1 ☐ 2 ☒ 5 ☐ 14 ☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20

Key Life: 86400 sec

Local ID: 232413

☒ Dead Peer Detection

☒ NAT Traversal

☐ Enable Local LAN

Phase 2

☐ Enable Local LAN

IKE Proposal: Encryption: AES128 Authentication: SHA1

Encryption: AES256 Authentication: SHA256

Key Life: ☒ 43200 Seconds ☐ 5120 KBytes

☒ Enable Replay Detection

☒ Enable Perfect Forward Secrecy (PFS)

DH Group: 5

Im Browser

IP-Adresse 10.40.1.41 aufrufen, "vSphere Client starten"

Credentials für vSphere-Client

Username: ot-aufzug@atropos.local

Passwort: =,gssA%8ZxWc}m'0H"N;

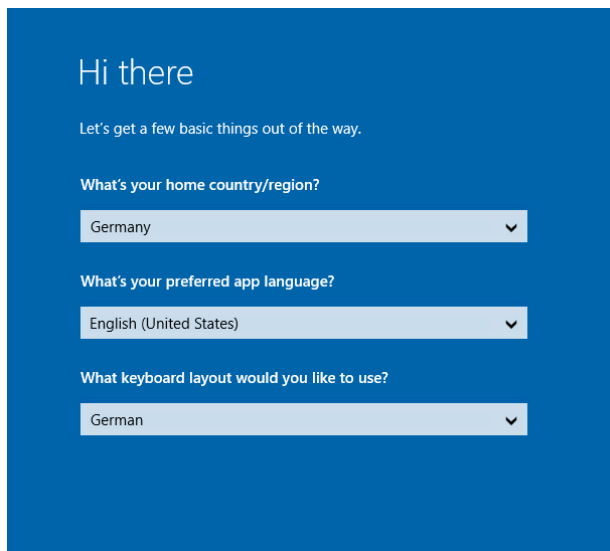
Fortigate Licenses

Aa Fortigate	License
 oslo-fw	FGVMSLTM26008268.lic
 th-fw	FGVMSLTM26008269.lic

[NX_Fortinet_Anleitung \(1\).docx](#)

Geräte

- Windows Server: 5
 - Administrator / Ganzgeheim123!
 - Preferred app language: English (US)



- Ubuntu Server: 3
 - junioradmin / junioradmin
- Windows Clients: 4
 - junioradmin / junioradmin
- Ubuntu Clients: 4
 - junioradmin / junioradmin
- Fortigate: 2
 - admin / Junioradmin123!
- pfSense: 1
 - admin / junioradmin
- C8000v: 1 für RADIUS, 2 für Backbone
 - cisco / cisco
- Grafana:1

- admin / junioradmin
- vSwitches: 5

Hostnames: So wie im Netzplan, ebenfalls klein geschrieben

Geräte

Geräte LBT

Name	Standort	Typ	Interface	IP 1	IP 2	IP 3	IP 4	Default Gateway	DNS
<u>bb1</u>	Backbone	Cisco IOS	bb1-bb2 bb1-scavenger	203.0.113.5/30	203.0.113.3/30				
<u>bb2</u>	Backbone	Cisco IOS	bb1-bb2 bb2-oslo-fw bb2-toronto	203.0.113.6/30	203.0.113.1/30	203.0.113.0/30			
<u>oslo-ansible</u>	Osolemio	Ubuntu	osolemio-mgmt-lan	10.3.3.201				10.3.2.254	
<u>oslo-ca</u>	Osolemio	Windows Server	osolemio-dc10.3.3.203					10.3.3.254	10.3.3.201
<u>oslo-dc1</u>	Osolemio	Windows Server	osolemio-dc10.3.3.201					10.3.3.254	10.3.3.201
<u>oslo-dc2</u>	Osolemio	Windows Server	osolemio-dc10.3.3.202					10.3.3.254	10.3.3.201
<u>oslo-fw</u>	Osolemio	FortiOS	bb2-oslo-fw osolemio-dmz osolemio-domain-lan osolemio-mgmt-lan	10.3.1.254	10.3.2.254	10.3.3.254	203.0.113.9/30		
<u>oslo-mail1</u>	Osolemio	Ubuntu	osolemio-dc10.3.3.101					10.3.3.254	10.3.1.102
<u>oslo-mail2</u>	Osolemio	Ubuntu	osolemio-dc10.3.3.102					10.3.3.254	10.3.1.102
<u>oslo-paw</u>	Osolemio	Windows 11	osolemio-mgmt-lan	10.3.3.201				10.3.2.254	
<u>oslo-r</u>	Osolemio	Cisco IOS	osolemio-dc10.3.3.205					10.3.3.254	10.3.3.201
<u>oslo-rds</u>	Osolemio	Windows Server	osolemio-dc10.3.3.204					10.3.3.254	10.3.3.201
<u>oslo-srv1</u>	Osolemio	Ubuntu	osolemio-dc10.3.1.101					10.3.1.254	10.3.1.102
<u>oslo-srv2</u>	Osolemio	Ubuntu	osolemio-dc10.3.1.102					10.3.1.254	10.3.1.102
<u>oslo-ws</u>	Osolemio	Windows 11	osolemio-dc10.3.3.206						
<u>sv-rodcc</u>	Scavenger	Windows Server	scavenger	10.1.0.201					
<u>sv-vpn</u>	Scavenger	pfSense	bb1-scavenger scavenger	10.1.0.254					
<u>sv-ws</u>	Scavenger	Windows 11	scavenger	10.1.0.X					
<u>th-dc1</u>	Toronto	Windows Server	toronto	10.2.0.201				10.2.0.254	10.3.3.201
<u>th-fw</u>	Toronto	FortiOS	bb2-toronto toronto	10.2.0.254	203.0.113.2/30				

Aa Name	≡ Standort	🕒 Typ	≡ Interface	≡ IP 1	≡ IP 2	≡ IP 3	≡ IP 4	≡ Default Gateway	≡ DNS
<u>th-srv</u>	Toronto	Ubuntu	toronto	10.2.0.202				10.2.0.254	10.3.3.201
<u>th-ws</u>	Toronto	Windows 11	toronto	10.2.0.X				10.2.0.254	

Allgemein

Subnetze: Alle Netzwerke haben eine SNM von /24, außer anders angegeben

DNS: Alle Geräte - bis auf die in der DMZ und oslo-dc1 & oslo-dc2 - haben oslo-dc1 (10.3.3.201) und oslo-dc2 (10.3.3.202) als DNS-Server konfiguriert.

AD DS

DSRM Passwort für alle DCs: Ganzgeheim123!

Oslo

- Domain: corp.equinor.no
- DCs:
 - oslo-dc1
 - oslo-dc2
- Clients (AD-gejoint):
 - oslo-ca
 - oslo-rds
 - oslo-ws
- Site:
 - Oslo
 - Subnet: 10.3.3.0/24
 - Site-Link "Oslo Stavanger" zwischen Oslo und Stavanger
 - Site-Link "Oslo Trondheim" zwischen Oslo und Trondheim
- DHCP:
 - Scope: 10.3.3.0/24
 - Excluded addresses: 10.3.3.201-205, 10.3.3.254
- DHCP Failover zwischen oslo-dc1 und oslo-dc2 mit Shared Secret "Ganzgeheim123!"

Trondheim

- Domain: remote.corp.equinor.no
- DCs:
 - th-dc1
- Clients (AD-gejoint):
 - th-ws
- Site:
 - Trondheim
 - Subnet: 10.2.0.0/24

- DHCP:
 - Scope: 10.2.0.0/24
 - Excluded addresses: 10.2.0.201-202, 10.2.0.254

Stavanger

- Domain: corp.equinor.no
- DCs:
 - sv-rodc (Read-Only Domain Controller = RODC)
- Clients (AD-gejoint):
 - sv-ws
- Site:
 - Stavanger
 - Subnet: 10.1.0.0/24
- DHCP:
 - Scope: 10.1.0.0/24
 - Excluded addresses: 10.1.0.201, 10.1.0.254

AD User

Benjamin Zwettler / zwettler / Ganzgeheim123!

Marcel Ogorevc / ogorevc / Ganzgeheim123!

Eduard Smola / smola / Ganzgeheim123!

AD CS

- Rolle AD CS installiert, Wizard konfiguriert
- Rolle Web Server (IIS) installiert und C:\CertEnroll geteilt, Berechtigungen auf C:\CertEnroll vergeben
- CDP- & AIA-Pfade konfiguriert, anscheinend fehlerhaft!
- Pfade ausgebessert → Reboot → pkiview.msc alles OK!
- Zertifikat "Workstation Authentication" für oslo-ws testweise erfolgreich ausgestellt

✓ Root-Zertifikat via GPO verteilen

✓ Auto-Enrollment via GPO einstellen

NPS/RADIUS

Rolle "Network Policy and Access Services" installieren auf **oslo-dc1** (siehe Netzplan)

Gruppe "Cisco Administrators" erstellen und User hinzufügen: zwettler, ogorevc, smola

Verbindung funktioniert auf den oslo-r (10.3.3.208) über telnet und ssh. Am oslo-dc1 wurde der telnet-Client installiert mit `Install-WindowsFeature telnet-client` und wird aufgerufen über folgenden Befehl: `telnet 10.3.3.208`, dann User-Eingabe und Passwort

1.5 Härtung

Baseline

ADMX-Dateien (Inhalt aus Ordner "Templates", inkl. en-US) aus dem Microsoft Security Compliance Toolkit (SCT) Download in `\\DEINE.DOMAIN\SYSVOL\DEINE.DOMAIN\Policies\PolicyDefinitions` kopiert - dadurch werden aber die bisher

gültigen GPO-Vorlagen aus `C:\Windows\PolicyDefinitions` überschreiben. Deshalb den Inhalt des Ordners `C:\Windows\PolicyDefinitions` in `\\DEINE.DOMAIN\SYSVOL\DEINE.DOMAIN\Policies\PolicyDefinitions` kopieren!

Gemini: 2. Das Problem: Der "Alles-oder-nichts"-Effekt

Sobald die GPMC erkennt, dass der Ordner `PolicyDefinitions` im SYSVOL existiert, schaltet sie komplett um. Sie ignoriert ab diesem Moment die lokalen Vorlagen auf deiner Festplatte (`C:\Windows\PolicyDefinitions`) und schaut **nur noch** in den Central Store.

Wenn du dort nur die Vorlagen aus der Windows Server 2022 Baseline reinkopiert hast, fehlen der GPMC alle Standard-Windows-Definitionen. Deshalb siehst du in alten GPOs plötzlich nichts mehr oder nur noch kryptische "Zusätzliche Registrierungseinstellungen".

Protected Users

zwettler, ogorevc, smola

Advanced Security Audit Policies

Gleiche Policies wie in Modul 3.3 gefordert umgesetzt. Testweise User "test" erstellt (Event-ID 4720) und gelöscht (Event-ID 4726)

Credential Guard

Auf **oslo-ws**

TPM-Modul in den Geräteeinstellungen hinzugefügt

In den VM-Optionen "Auf Virtualisierung basierende Sicherheit" aktiviert

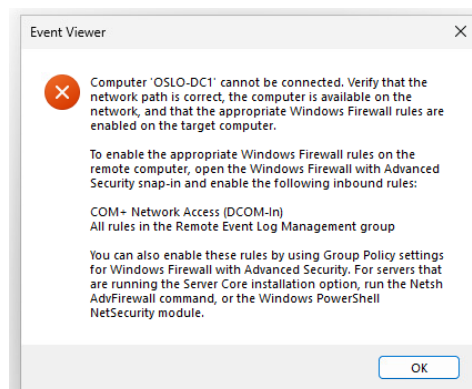
GPO erstellt, Aktivierung von **"Computerkonfiguration > Administrative Vorlagen > System > Device Guard"**

DFS - Module 4.2 A + B & Modul 4.3

Modul 4.2 - Implementieren von DFS

Modul 4.3 - File Service Security

Jump-Host



Auf Windows-Servern: Firewall

✓ Cast to Device streaming server (RTSP-Streaming-In)	Cast to Device functionality	Public	Yes
✓ Cast to Device UPnP Events (TCP-In)	Cast to Device functionality	Public	Yes
✓ COM+ Network Access (DCOM-In)	COM+ Network Access	All	Yes
COM+ Remote Administration (DCOM-In)	COM+ Remote Administration	All	No
✓ Core Networking - Destination Unreachable (ICMPv6-In)	Core Networking	All	Yes
✓ Core Networking - Destination Unreachable Fragmentation (ICMPv6-In)	Core Networking	All	Yes

Network Discovery (WSD-In)	Network Discovery	All	No	Allow	No
✓ OpenSSH SSH Server (sshd)	OpenSSH Server	All	Yes	Allow	No
✓ Performance Logs and Alerts (DCOM-In)	Performance Logs and Alerts	Private...	Yes	Allow	No
✓ Performance Logs and Alerts (DCOM-In)	Performance Logs and Alerts	Domain	Yes	Allow	No
✓ Performance Logs and Alerts (TCP-In)	Performance Logs and Alerts	Domain	Yes	Allow	No
✓ Performance Logs and Alerts (TCP-In)	Performance Logs and Alerts	Private...	Yes	Allow	No
Remote Desktop - Shadow (TCP-In)	Remote Desktop	All	No	Allow	No
✓ Remote Desktop - User Mode (TCP-In)	Remote Desktop	All	Yes	Allow	No
Remote Desktop - User Mode (UDP-In)	Remote Desktop	All	No	Allow	No
Remote Desktop - (TCP-WS-In)	Remote Desktop (WebSocket)	All	No	Allow	No
Remote Desktop - (TCP-WSS-In)	Remote Desktop (WebSocket)	All	No	Allow	No
✓ Remote Event Log Management (NP-In)	Remote Event Log Management	All	Yes	Allow	No
✓ Remote Event Log Management (RPC)	Remote Event Log Management	All	Yes	Allow	No
✓ Remote Event Log Management (RPC-EPMAP)	Remote Event Log Management	All	Yes	Allow	No
Remote Event Monitor (RPC)	Remote Event Monitor	All	No	Allow	No
Remote Event Monitor (RPC-EPMAP)	Remote Event Monitor	All	No	Allow	No
✓ Remote Scheduled Tasks Management (RPC)	Remote Scheduled Tasks Management	All	Yes	Allow	No
✓ Remote Scheduled Tasks Management (RPC-EPMAP)	Remote Scheduled Tasks Management	All	Yes	Allow	No
✓ Remote Service Management (NP-In)	Remote Service Management	All	Yes	Allow	No
✓ Remote Service Management (RPC)	Remote Service Management	All	Yes	Allow	No
✓ Remote Service Management (RPC-EPMAP)	Remote Service Management	All	Yes	Allow	No
Inbound Rule for Remote Shutdown (RPC-EP-In)	Remote Shutdown	All	No	Allow	No
Inbound Rule for Remote Shutdown (TCP-In)	Remote Shutdown	All	No	Allow	No
✓ Remote Volume Management - Virtual Disk Service (RPC)	Remote Volume Management	All	Yes	Allow	No
✓ Remote Volume Management - Virtual Disk Service Loadable (RPC)	Remote Volume Management	All	Yes	Allow	No
✓ Remote Volume Management (RPC-EPMAP)	Remote Volume Management	All	Yes	Allow	No
Routing and Remote Access (GRE-In)	Routing and Remote Access	All	No	Allow	No
Routing and Remote Access (L2TP-In)	Routing and Remote Access	All	No	Allow	No
Routing and Remote Access (PPTP-In)	Routing and Remote Access	All	No	Allow	No
Windows Defender Firewall Remote Management (RPC)	Windows Defender Firewall Remote Management	All			
Windows Defender Firewall Remote Management (RPC)	Windows Defender Firewall Remote Management	All			
✓ Windows Management Instrumentation (ASync-In)	Windows Management Instrumentation (WMI)	All			
✓ Windows Management Instrumentation (DCOM-In)	Windows Management Instrumentation (WMI)	All			
✓ Windows Management Instrumentation (WMI-In)	Windows Management Instrumentation (WMI)	All			
Windows Media Player (UDP-In)	Windows Media Player	All			

```
# Testet, ob du Hardware-Infos via WMI/CIM (modern) bekommst
Get-CimInstance -ComputerName oslo-ca -ClassName Win32_PnPEntity | Select-Object -First 5
```

Device-Manager funktioniert nicht, da deprecated und somit nur schlecht unterstützt

Auf dem Desktop von User CORP\Administrator auf oslo-paw liegt die MMC-Datei.

Achtung! Bei einer SSH-Verbindung auf oslo-paw muss "CORP\" dem Usernamen vorangestellt werden, also z.B.

```
ssh CORP\Administrator@oslo-paw
```

ÜBERTRÄGUNG NACH VM

https://htl3r.sharepoint.com/sites/2425_4CN_NWT_SDO_KUS/Freigegebene%20Dokumente/General/Unterlagen-KUS/Literatur_Referenzen/Linux-Server-%20Das%20umfassende%20Handbuch%20--%20Dirk%20Deimeke;%20Daniel%20van%20Soest;%20Stefan%20Kania;%20Peer%20Heinlein;%20--%207,%202023%20--%20Rheinwerk%20Verlag%20GmbH.pdf?CT=1775307619953&OR=ItemsView&wdOrigin=TEAMSFILE.FILEBROWSER.DOCUMENTLIBRARY

root-ca.crt

root-ca.pem_compressed.zip

l.zip

```
sudo openssl x509 -inform DER -in /etc/ssl/certs/root-ca.crt -out /etc/ssl/certs/root-ca.pem
```

Edit /etc/dovecot/conf.d/10-ssl.conf

```
ssl_cert = </etc/ssl/certs/oslo-srv2-bundle.crt
```

```
ssl_key = </etc/ssl/private/oslo-srv2.key
```

```
[req]
distinguished_name = req_distinguished_name
req_extensions = v3_req
prompt = no

[req_distinguished_name]
C = NO
L = Oslo
O = Equinor
CN = oslo-srv2.corp.equinor.no

[v3_req]
keyUsage = digitalSignature, keyEncipherment
extendedKeyUsage = serverAuth
subjectAltName = @alt_names

[alt_names]
DNS.1 = oslo-srv2.corp.equinor.no
DNS.2 = mail.corp.equinor.no
DNS.3 = smtp.corp.equinor.no
DNS.4 = imap.corp.equinor.no
DNS.5 = pop3.corp.equinor.no

certreq -submit -attrib "CertificateTemplate:OsloMailServer" Downloads\oslo-srv2.csr

Datei: /etc/dovecot/conf.d/10-mail.conf
mail_location = maildir:~/Maildir

Datei: /etc/dovecot/conf.d/10-master.conf
service auth{
    ...
    # Postfix smtp-auth
    unix_listener /var/spool/postfix/private/auth {
        mode = 0660
        user = postfix
        group = postfix
    }
}
```

```
Datei: /etc/dovecot/conf.d/10-ssl.conf
ssl = required
ssl_cert = </etc/ssl/certs/oslo-srv2.crt
ssl_key = </etc/ssl/private/oslo-srv2.key
# Optional (falls erstellt):
# ssl_ca = </etc/ssl/certs/root-ca.crt

# Convert DER/Binary to PEM/Text
sudo openssl x509 -inform DER -in /etc/ssl/certs/root-ca.crt -out /etc/ssl/certs/root
-ca.pem

# Create the bundle using the PEM version
sudo cat /etc/ssl/certs/oslo-srv2.crt /etc/ssl/certs/root-ca.pem | sudo tee /etc/ssl/
certs/oslo-srv2-bundle.crt
```